

Trust-Based Detection and Mitigation of Poisoning Attacks in Swarm Learning

Using Participant Contribution and Model Behavior

Chapter 1: Introduction and Background

1. Proposed Research Title

- **Primary Title:** Trust-Based Detection and Mitigation of Poisoning Attacks in Swarm Learning Using Participant Contribution and Model Behavior
- **Alternative Title:** An Adaptive Multi-Factor Trust Mechanism for Poisoning-Resistant Swarm Learning

2. Research Background

- Swarm Learning enables multiple participants to collaboratively train a machine learning model while keeping their local training data private.
- Each participant trains a model using its own local dataset and shares model information rather than the original data.
- The shared model information is aggregated to create a global model, which is then used in subsequent training rounds.
- This architecture is highly useful when organizations cannot directly share their data due to privacy or regulatory constraints.
- However, because participants contribute model updates independently, a malicious participant can manipulate its local training process or model update to intentionally degrade the performance of the global model.
- This vulnerability is commonly referred to as a poisoning attack.
- Recent work has demonstrated that Swarm Learning is vulnerable to poisoning attacks and has proposed Byzantine-resilient defenses (e.g., a 2024 study proposed a privacy-preserving Byzantine-resilient mechanism for e-healthcare).
- Our project will not focus on specific domains like medical diagnosis. Instead, it will investigate the general problem of detecting and mitigating malicious participants in Swarm Learning.

3. Problem Statement

- A straightforward way to detect malicious participants is to compare their model updates with other participants' updates.
- For example, if Participant 3 produces an update (U3) that is very different from Participant 1 (U1) and Participant 2 (U2), the system may consider Participant 3 suspicious.
- However, this approach has an important flaw: different does not necessarily mean malicious.

- Suppose Participant 1 has mostly Class A data, Participant 2 has mostly Class B data, and Participant 3 has mostly Class C data.
- Participant 3 may produce an update that is significantly different from the others even though they are completely honest, simply due to heterogeneous (non-IID) data distributions.
- Research on Byzantine-robust learning has similarly highlighted the difficulty of distinguishing malicious outliers from legitimate updates under non-IID data.
- Therefore, low similarity should not automatically mean a participant is malicious.
- Similarly, high similarity should not automatically mean honest, as a malicious participant could potentially craft an update that appears similar to legitimate updates to evade detection.

Chapter 2: Research Gap and Proposed Concept

4. Main Research Gap

- The main research gap we want to investigate is whether the actual contribution of a participant's update to the global model can be used, together with other behavioral indicators, to improve poisoning detection.
- This approach aims to reduce the incorrect rejection of honest participants who simply possess heterogeneous data.
- Instead of only asking, "Does this update look different?", the system asks, "What effect does this update have on the global model?"
- This shifts the research angle from simple update anomaly detection to contribution- and behavior-based trust evaluation.

5. Core Research Idea

- Our main idea is to evaluate every participant's update before final aggregation.
- Instead of immediately aggregating all updates (e.g., Update 1, 2, and 3 from Participants 1, 2, and 3), the system evaluates their individual contribution.
- The system can compare the global performance using all updates against the performance when one update is omitted.
- The difference in performance provides an estimate of the omitted participant's contribution, which can be repeated for every participant.

6. Leave-One-Out Contribution

The main mechanism we propose is Leave-One-Out (LOO) contribution evaluation.

Example Validation Process:

All participants (P1 + P2 + P3) → Aggregation → Validation → Accuracy = **90%**

Evaluating P1:

Remove P1 (P2 + P3) → Aggregation → Validation → Accuracy = **86%**

Contribution(P1) = 90% - 86% = +4%

P1 appears to be contributing positively.

Evaluating P2:

Remove P2 (P1 + P3) → Aggregation → Validation → Accuracy = **94%**

Contribution(P2) = 90% - 94% = -4%

Removing P2 improves the model, making P2 suspicious.

However, we will NOT immediately classify P2 as malicious based on this single metric.

7. Why Leave-One-Out Alone Is Not Enough

- An honest participant with a very different data distribution may sometimes have a negative contribution to the global model.
- For example, if P1 and P2 have normal data but P3 has normal yet highly non-IID data, P3 could temporarily reduce global validation performance without being malicious.
- Therefore, a negative contribution does not automatically equal malicious intent.
- This limitation is exactly why our proposed solution utilizes multiple factors to build a holistic trust score.

Chapter 3: The Multi-Factor Trust Mechanism

8. Proposed Multi-Factor Trust Mechanism

Each participant receives a trust score based on multiple observations. The proposed factors are:

Factor 1 — Leave-One-Out Contribution

- Measures how the global model performs when the specific participant's update is removed.
- This serves as the primary factor in determining the practical utility of the update.

Factor 2 — Local Validation Performance

- The participant evaluates its locally trained model on its own validation data.
- This helps determine whether the update is consistent with useful local learning before it is shared globally.

Factor 3 — Update Magnitude

- The system checks whether the participant's update is unusually large compared with other updates.
- An unusually large update (e.g., P3 having a massive gradient compared to P1 and P2) can be treated as a suspicious signal indicative of tampering.

Factor 4 — Update Similarity

- Cosine similarity is still calculated but acts as supporting evidence rather than the final decision mechanism.
- High similarity combined with a positive contribution provides strong evidence of normal behavior.
- Low similarity combined with a negative contribution provides stronger suspicion of an attack.
- Crucially, low similarity alone is NOT classified as malicious.

Factor 5 — Historical Behavior

- The system maintains a record of the participant's previous behavior across training rounds.
- A participant exhibiting fluctuating contributions (e.g., positive, then negative, then positive) may simply have heterogeneous data.
- Conversely, a participant consistently producing harmful updates round after round provides strong evidence of malicious behavior.
- This allows trust to be updated and stabilized over time.

9. Proposed Trust Score

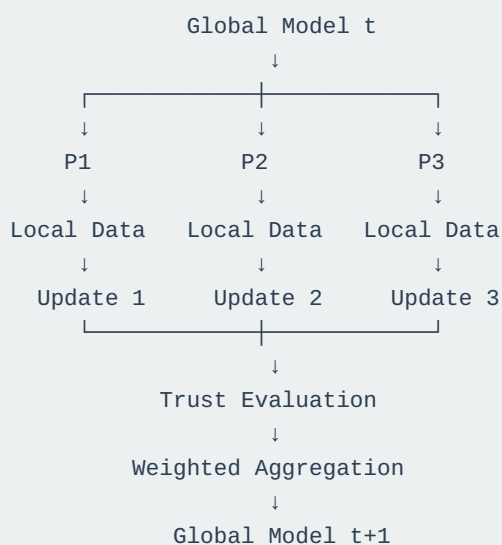
A simplified formulation of the trust calculation:

$$\text{Trust}(P_i) = w_1 \times \text{ContributionScore} + w_2 \times \text{ValidationScore} + w_3 \times \text{MagnitudeScore} + w_4 \times \text{SimilarityScore} + w_5 \times \text{HistoricalScore}$$

where $w_1 + w_2 + w_3 + w_4 + w_5 = 1$

The exact weighting should be determined experimentally rather than arbitrarily fixed.

10. How Aggregation Will Work



- The important point is that the original data never needs to be combined or shared.

- Each participant keeps its local dataset strictly private.
- Only model updates are evaluated, scored for trust, and subsequently aggregated based on those weights.

11. Example of the Proposed System

- Assume there are five participants, where P4 is designated as an attacker.
- **Round 1:** All honest nodes produce normal updates, while P4 produces a poisoned update.
- The system evaluates all updates and generates trust scores (e.g., P1, P2, P3, P5 score ~ 0.90 , while P4 scores 0.35).
- P4 is suspicious, but the system does not immediately remove it; instead, its update receives a much lower aggregation weight.
- **Round 2:** P4 produces another harmful update. The system observes the negative contribution again, and historical trust drops further (e.g., P4 Trust = 0.22).
- **Round 3:** P4 continues the attack. Trust drops to 0.10, and its update now receives almost no influence on the global model.
- This phased approach allows the system to react robustly to repeated malicious behavior rather than making brittle decisions based on a single unusual update.

Chapter 4: Methodology and Experiments

12. How We Will Create Malicious Participants

- We do not need naturally occurring malicious datasets. For experimental purposes, we will create controlled poisoning attacks using a standard machine-learning dataset.
- **Label-Flipping Attack:** The attacker intentionally swaps labels in their local dataset (e.g., changing Class A to Class B). The participant trains using this poisoned data and sends the resulting harmful model update.
- **Sign-Flipping Attack:** The attacker directly manipulates the direction of the computed model update. Conceptually, if a normal update is ΔW , the malicious update is crafted as $-\Delta W$. The attack strength can also be scaled.
- **Backdoor Attack:** The attacker trains the model to behave normally on regular samples but produce a targeted misclassification when a specific trigger is present. We will initially avoid this complexity and begin with label-flipping and sign-flipping.

13. Handling Non-IID Data

- Handling heterogeneous data is one of the most critical aspects of our research, requiring two distinct experimental conditions.
- **IID Condition:** Participants have approximately similar data distributions (e.g., balanced datasets across all nodes).
- **Non-IID Condition:** Participants have skewed or highly specialized data distributions (e.g., P1 holds mostly Class A, P2 mostly Class B).

- The purpose of these conditions is to determine whether the proposed mechanism incorrectly penalizes honest participants simply because their local data distributions differ from the global average.
- This distinction is important because traditional robust aggregation methods often struggle to differentiate between honest non-IID updates and active poisoning attacks.

14. Experimental Scenarios

The project will be evaluated using four major scenarios:

1. **Scenario 1 — Normal Swarm Learning:** All participants are honest. This establishes the baseline performance of the architecture.
2. **Scenario 2 — Honest Non-IID Swarm Learning:** All participants remain honest but possess skewed data distributions. This measures the false positive rate of the trust mechanism on legitimate heterogeneous nodes.
3. **Scenario 3 — Swarm Learning Under Poisoning:** Attacks are introduced without the proposed defense. This measures the maximum potential damage of the attacks.
4. **Scenario 4 — Swarm Learning With Proposed Defense:** Attacks are introduced with the multi-factor trust mechanism enabled. This evaluates the system's defensive efficacy.

15. Evaluation Metrics

- **Global Accuracy:** Measures the final global model's predictive performance on test data.
- **F1 Score:** Measures the balance between precision and recall, especially useful for imbalanced datasets.
- **Attack Success Rate:** Measures how successfully the attacker achieves its poisoning objective (e.g., the targeted misclassification rate).
- **False Positive Rate:** Measures how often completely honest participants are incorrectly classified as suspicious or malicious.
- **Detection Rate:** Measures how many actively malicious participants are successfully detected and mitigated by the system.
- **Convergence:** Measures how quickly (in terms of communication rounds) the global model reaches stable, optimal performance.

16. Baseline Comparisons

- To demonstrate the utility of the proposed approach, it will be compared against simpler baseline methods rather than being tested in isolation.
- We will evaluate: Standard FedAvg (no defense) vs. Similarity-based detection vs. Contribution-based detection vs. Our Proposed Multi-Factor Trust.
- This comparative analysis will definitively answer whether combining multiple behavioral factors provides a measurable advantage over single-signal defenses.
- Recent literature on flexible federated learning supports the relevance of analyzing participant contribution under both IID and non-IID settings.

Chapter 5: Project Scope and Contributions

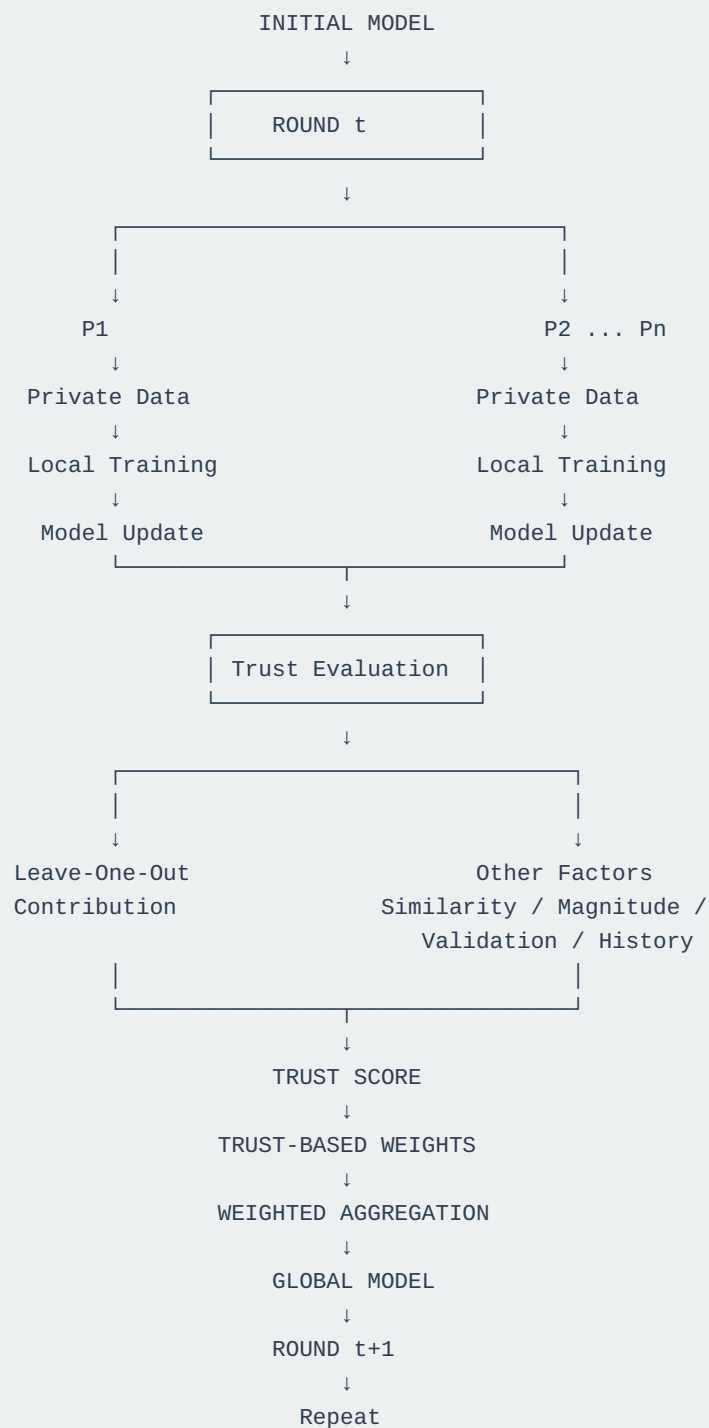
17. Important Research Limitation

- We must explicitly acknowledge that an isolated contribution score is not automatically trustworthy.
- Recent research indicates that sophisticated attackers can manipulate their model updates to receive artificially high contribution scores, tricking single-signal defenses.
- Therefore, our project does NOT claim that Leave-One-Out evaluation perfectly or flawlessly identifies malicious participants.
- Instead, the core research question centers on whether combining contribution-based evaluation with secondary behavioral signals creates a more robust defense than relying on any single metric.
- This framing ensures the research is scientifically realistic and defensible.

18. Expected Research Contribution

- The primary expected contribution is a comprehensive multi-factor trust-based defense mechanism for Swarm Learning environments.
- The system evaluates individual updates, measures leave-one-out contribution, examines validation and magnitude, and uses similarity as supporting context.
- By maintaining historical behavior logs, it generates dynamic trust scores used to assign weighted influence during aggregation.
- The framework will provide empirical evidence reducing the influence of suspicious updates while carefully preserving the input of honest participants in complex non-IID scenarios.

19. Overall Architecture



20. Final Research Angle

- The project goes beyond the narrow claim of "detecting malicious participants using cosine similarity."
- Instead, the true angle is: "We investigate whether participant contribution and model behavior can be combined into an adaptive trust mechanism to detect and mitigate poisoning attacks in Swarm Learning, while reducing false detection of honest participants caused by heterogeneous/non-IID data."
- It pivots the defensive logic from the traditional "Does the update look unusual?" to the functional "What does this update actually do to the global model?"

21. Scope for the Project

- To maintain an achievable scope, the first implementation will utilize a Swarm Learning simulation involving 3 to 10 participants.
- Experiments will run on one standard classification dataset, tested across both IID and non-IID splits over 20 to 50 communication rounds.
- The evaluation will initially feature 1 or 2 straightforward poisoning attack types (e.g., label flipping).
- The focus remains on implementing the Leave-One-Out contribution, multi-factor trust scoring, and trust-weighted aggregation.
- We will intentionally avoid compounding complexities like blockchain integration or heavy cryptographic overheads, as the core research value lies in the behavioral trust mechanism itself.

22. One-Line Explanation for the Project Guide

“Our research proposes a multi-factor trust mechanism for Swarm Learning that evaluates how each participant's update contributes to the global model, combines this with validation, update behavior, similarity, and historical evidence, and uses the resulting trust score to reduce the influence of poisoning participants while preserving honest participants with heterogeneous data.”